

Supplementary Information for: Ten Edge-Degree Counts Are All a Degree-Based Topological Index Can See

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S1 The baseline

Table S1: The 30-index classical baseline.

Family	Indices
Degree-based, BID (18)	$M_1, M_2, mM_1, mM_2, F, R, \text{SCI}, H, GA$ [?], AG, ABC, ABS, AZI, SO [?], SO_{red}, Alb [?], $\sigma, \text{red}M_1$
Distance-based (8)	$W, WW, HR, \text{degree distance}, \text{Gutman}, \text{Mostar}, Sz, PI$
Spectral (3)	Estrada EE, energy, spectral radius
Other (1)	Balaban J

S2 The value matrix of the baseline

S3 Definitions of the 20 alternative candidate indices

For reproducibility we give the formal definition of each of the twenty candidate indices screened in Section ???. Let $G = (V, E)$ be a simple connected hydrogen-suppressed molecular graph with $n = |V|$ vertices, $m = |E|$ edges, degree sequence $\{d_v\}_{v \in V}$, geodesic distance matrix $D = (d_{uv})$, adjacency matrix A , eccentricity $\varepsilon(v) = \max_u d_{uv}$, and adjacency-spectrum / Laplacian-spectrum eigenvalues $\{\lambda_i\} / \{\mu_i\}$ respectively. All twenty definitions are implemented in `src/novel_candidates.py` of the companion repository.

Information-theoretic (5). With $H(\{c_i\}) = -\sum_i (c_i / \sum_j c_j) \log(c_i / \sum_j c_j)$ the Shannon entropy of a count distribution:

$$\begin{aligned}
 \text{InfoH_deg}(G) &= H(\{|\{v : d_v = k\}|\}_k), \\
 \text{InfoH_dist}(G) &= H(\{|\{(u, v) : d_{uv} = k, u < v\}|\}_k), \\
 \text{InfoH_spec}(G) &= -\sum_i p_i \log p_i, \quad p_i = \lambda_i^2 / \sum_j \lambda_j^2, \\
 \text{InfoH_ecc}(G) &= H(\{|\{v : \varepsilon(v) = k\}|\}_k), \\
 \text{Bonchev_Id}(G) &= W \log_2 W - \sum_{k \geq 1} k d_k \log_2(k d_k),
 \end{aligned}$$

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Table S2: The value matrix Φ_B of the 18 degree-based baseline indices on the ten degree pairs $(i, j) \in P_4$: entry $f_b(i, j)$, obtained by evaluating each index on $K_{i,j}$ and dividing by its ij edges. Vertex-degree sums $\sum_v g(d_v)$ are written as $f(a, b) = g(a)/a + g(b)/b$. The matrix has rank 10 (smallest/largest singular value $1.2e - 04$).

Index	$f(a, b)$	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(2, 2)	(2, 3)	(2, 4)	(3, 3)	(3, 4)	(4, 4)
M_1	$a + b$	2	3	4	5	4	5	6	6	7	8
M_2	ab	1	2	3	4	4	6	8	9	12	16
mM_1	$a^{-3} + b^{-3}$	2	1.12	1.04	1.02	0.25	0.162	0.141	0.0741	0.0527	0.0312
mM_2	$1/(ab)$	1	0.5	0.333	0.25	0.25	0.167	0.125	0.111	0.0833	0.0625
F	$a^2 + b^2$	2	5	10	17	8	13	20	18	25	32
R	$(ab)^{-1/2}$	1	0.707	0.577	0.5	0.5	0.408	0.354	0.333	0.289	0.25
SCI	$(a + b)^{-1/2}$	0.707	0.577	0.5	0.447	0.5	0.447	0.408	0.408	0.378	0.354
H	$2/(a + b)$	1	0.667	0.5	0.4	0.5	0.4	0.333	0.333	0.286	0.25
GA	$2\sqrt{ab}/(a + b)$	1	0.943	0.866	0.8	1	0.98	0.943	1	0.99	1
AG	$(a + b)/(2\sqrt{ab})$	1	1.06	1.15	1.25	1	1.02	1.06	1	1.01	1
ABC	$\sqrt{(a + b - 2)/(ab)}$	0	0.707	0.816	0.866	0.707	0.707	0.707	0.667	0.645	0.612
ABS	$\sqrt{(a + b - 2)/(a + b)}$	0	0.577	0.707	0.775	0.707	0.775	0.816	0.816	0.845	0.866
AZI	$(ab/(a + b - 2))^3$ (0 on K_2)	0	8	3.38	2.37	8	8	8	11.4	13.8	19
SO	$\sqrt{a^2 + b^2}$	1.41	2.24	3.16	4.12	2.83	3.61	4.47	4.24	5	5.66
SO _{red}	$\sqrt{(a - 1)^2 + (b - 1)^2}$	0	1	2	3	1.41	2.24	3.16	2.83	3.61	4.24
Alb	$ a - b $	0	1	2	3	0	1	2	0	1	0
σ	$(a - b)^2$	0	1	4	9	0	1	4	0	1	0
redM ₁	$(a - 1)^2/a + (b - 1)^2/b$	0	0.5	1.33	2.25	1	1.83	2.75	2.67	3.58	4.5

where in the last line $d_k = |\{(u, v) : d_{uv} = k, u < v\}|$ and $W = \sum_k k d_k$ is the Wiener index; this Bonchev-type form, as implemented, differs from the standard information-theoretic Wiener index $I_D^W = W \log_2 W - \sum_k k d_k \log_2 k$.

Centrality-based (5). With NetworkX centralities (betweenness normalised by $\binom{n-1}{2}$, NetworkX's default):

$$\begin{aligned}
\text{Btw_sum}(G) &= \sum_{v \in V} c_B(v), \quad c_B(v) = \binom{n-1}{2}^{-1} \sum_{s \neq t \neq v} \frac{\sigma_{st}(v)}{\sigma_{st}}, \\
\text{Cls_sum}(G) &= \sum_v (n-1) \left(\sum_{u \neq v} d_{uv} \right)^{-1}, \\
\text{Eig_sum}(G) &= \sum_v x_v, \quad x \text{ the (positive) Perron eigenvector of } A, \\
\text{Harm_cent}(G) &= \sum_v \sum_{u \neq v} \frac{1}{d_{uv}} = 2 \text{HR}(G), \\
\text{Btw_var}(G) &= \text{Var}_v(c_B(v)).
\end{aligned}$$

Spectral-hybrid (3). With $\text{diam}(G) = \max_{u,v} d_{uv}$ and Estrada index $\text{EE}(G) = \sum_i e^{\lambda_i}$:

$$\begin{aligned}
\text{AlgConn}(G) &= \mu_2(G), \quad \text{second-smallest Laplacian eigenvalue,} \\
\text{Rho} \times \text{Dia}(G) &= \rho(G) \cdot \text{diam}(G), \quad \rho(G) = \max_i |\lambda_i|, \\
\text{EE} \times \text{W}(G) &= \log(1 + \text{EE}(G)) \cdot \log(1 + W(G)).
\end{aligned}$$

Motif / structural (4).

$$\begin{aligned}\text{Triangles}(G) &= \frac{1}{6} \text{tr}(A^3), \\ \text{MeanClust}(G) &= \frac{1}{n} \sum_v \frac{2t(v)}{d_v(d_v - 1)}, \quad t(v) = \text{triangles at } v, \\ \text{FourCyc}(G) &= \frac{1}{8} (\text{tr}(A^4) - 2 \sum_v d_v(d_v - 1) - 2m), \\ \text{CutVerts}(G) &= |\{v \in V : G - v \text{ is disconnected}\}|.\end{aligned}$$

Eccentricity-based (3).

$$\begin{aligned}\text{TotEcc}(G) &= \sum_{v \in V} \varepsilon(v), \\ \text{Radius}(G) &= \min_{v \in V} \varepsilon(v), \\ \text{MeanEcc}(G) &= \frac{1}{n} \sum_{v \in V} \varepsilon(v).\end{aligned}$$

None of these twenty candidates is claimed as a newly invented index. The information-theoretic family follows the information-theoretic descriptor tradition [?]; the centrality family follows the standard graph-theory literature; the motif and eccentricity families are textbook graph invariants; the spectral-hybrid combinations are deliberately constructed novel pairings to test whether non-standard product indices escape the redundancy ceiling. They appear here purely as screening targets to test the generality of the target-aware screen, and the verdict reported in Section ?? is a statement about this particular candidate set under the tested baselines and thresholds, not a claim that any individual candidate is or is not a useful QSPR descriptor.

S4 Statistical details of the benchmark

The benchmark of the main text uses $r = 5$ repeats of $k = 5$ -fold cross-validation with folds grouped by isomorphism class of the hydrogen-suppressed graph. For a paired difference d_{ij} (repeat i , fold j) between two models, the Nadeau–Bengio corrected variance of the mean difference is $\hat{\sigma}^2 (1/(kr) + n_{\text{test}}/n_{\text{train}})$, where $\hat{\sigma}^2$ is the sample variance of the kr differences; the corrected t statistic has $kr - 1$ degrees of freedom and the reported 90% intervals are $\bar{d} \pm t_{0.95, kr-1} \hat{\sigma}_{\text{corr}}$. Equivalence is tested by two one-sided tests (TOST) at level 0.05 against a margin of 5% of the reference model’s mean RMSE, or 0.05 ($\text{AUC}_{\text{ref}} - 0.5$) for BBBP; equivalence is declared when the 90% interval lies inside $[-\delta, \delta]$. Within each dataset the five comparisons are Holm-adjusted. The inner cross-validation that selects the LASSO or ℓ_1 -logistic penalty is also grouped by graph. Random forests use 300 trees.

Screened subsets and the combined screen. Restricting the baseline to its pairwise-screened subset (about 6–8 indices) is equivalent within 5% for the random forest on the three regression sets and slightly better on BBBP; for LASSO it is equivalent within the margin on the regression sets, although the cost is distinguishable from zero on ESOL (+0.045) and Lipophilicity (+0.009), and it is not shown equivalent on BBBP. The combined screen, which also requires $|\text{pcor}| \geq 0.10$, is a diagnostic and not a feature selector: on Lipophilicity it removes every feature in 24 of 25 folds.

S5 Datasets used for the generality check

The nine additional MoleculeNet files (BACE, ClinTox, SIDER, Tox21, ToxCast, HIV, MUV, QM8, QM9) are downloaded from the DeepChem mirror by script 78 of the repository; their SHA-256 digests are recorded in the results file `certificate_generality.csv`. Only the SMILES column of each file is used.